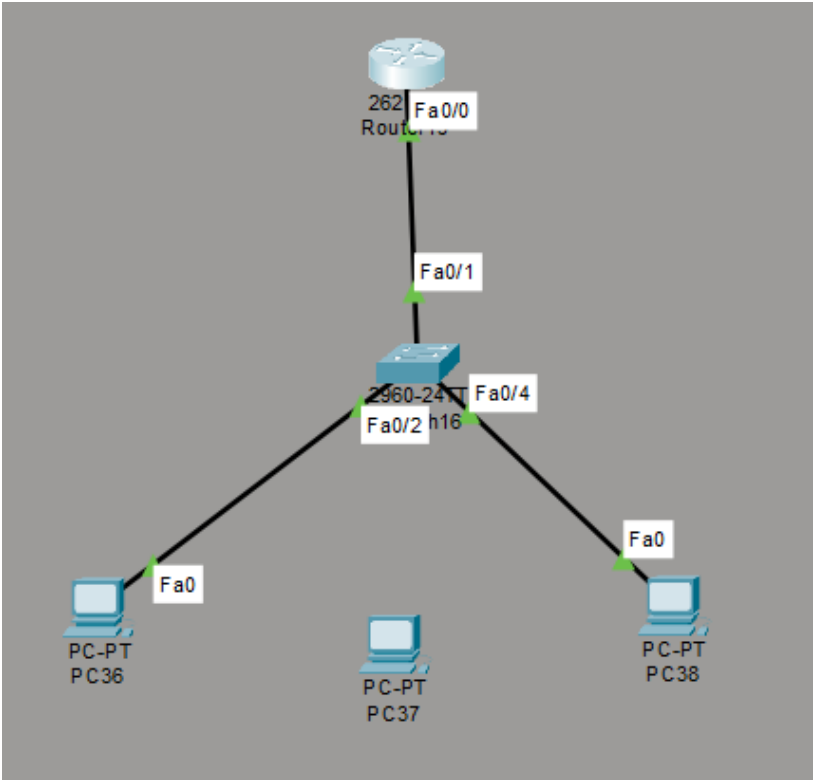


Q 1-List 5 common hardware issues that can cause network problems in a typical office setup, and for each, describe one symptom that helps identify it.

#	Hardware Issue	Common Symptom
1	Faulty Ethernet cable	Connection keeps disconnecting/reconnecting or shows no network connection.
2	Damaged/failed network card (NIC)	The computer cannot connect to the network even when the cable and switch port are working.
3	Faulty switch port	Devices connected to one particular port have no network access , while other ports work normally.
4	Failing router	Multiple users lose Internet access or cannot communicate with other networks.
5	Bad Wi-Fi access point	Users experience weak/intermittent Wi-Fi or cannot connect to the wireless network.

Q 2-Open Cisco Packet Tracer or any network simulator, set up a small network with 3 PCs, 1 switch, and 1 router, and intentionally disconnect one device. Use the simulator's troubleshooting tools to identify which device is faulty and document the steps you took.



```

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip
Router#show ip interface brief

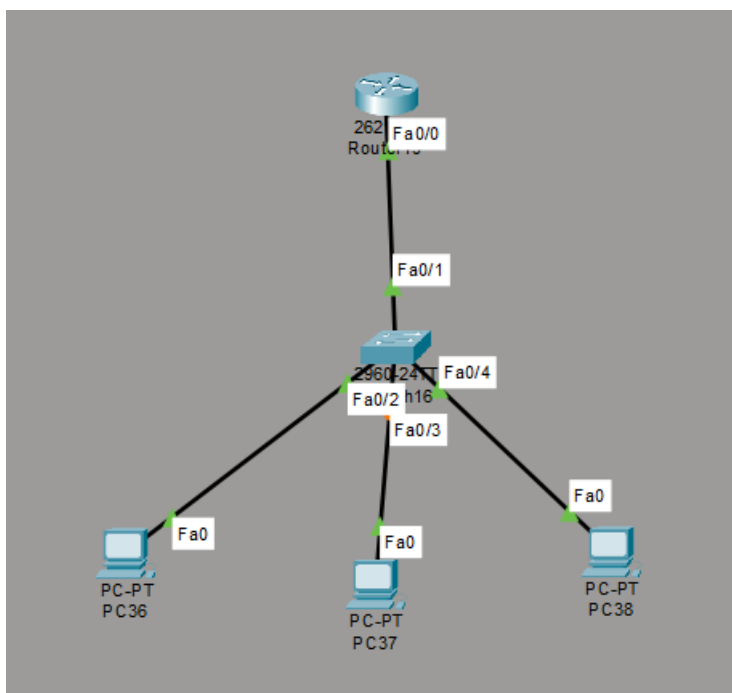
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.10.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down

```

Router#

```



Troubleshooting steps:

1. Checked the physical topology.
2. Observed that PC1's switch connection was down.
3. Sent a ping from PC0 to PC1.
4. Ping failed with Request timed out.
5. Tested PC2 and confirmed that the rest of the network was working.
6. Used Packet Tracer Simulation Mode to observe the failed ICMP communication.
7. Reconnected PC1's cable.
8. Tested the connection again.
9. Ping succeeded.

Fault: PC1 Ethernet cable was intentionally disconnected.

Symptom: PC1 could not communicate with other devices.

**Q 3- Create a troubleshooting checklist for diagnosing slow internet issues in a home WiFi network, including at least 8 steps covering both hardware and software checks.

Hint: Think about checking cables, router status lights, device settings, and running ping or speed tests.**

1. Check cables – Make sure the router's power cable and Ethernet/Internet cable are firmly connected and not damaged.

2. Check router status lights – Look for unusual red/orange lights or a missing Internet/WAN light.

3. Restart the router – Turn off the router, wait about 30 seconds, and turn it back on.

4. Check Wi-Fi signal strength – Move closer to the router and see whether the connection becomes faster. Walls and distance can weaken Wi-Fi.

5. Test another device – Check whether the Internet is slow on only one device or on multiple devices. If only one device is affected, the problem may be its settings or hardware.

6. Check device Wi-Fi settings – Make sure Wi-Fi is enabled, the correct network is selected, and Airplane Mode is disabled.

7. Run a ping test – Open Command Prompt/Terminal and try:

ping 8.8.8.8

High response times or packet loss can indicate a connectivity problem.

8. Run an Internet speed test – Compare the download/upload speed with the speed promised by your ISP. A large difference may indicate an ISP or network problem.

9. Check background usage – Pause downloads, cloud backups, streaming, or software updates that may be consuming bandwidth.

10. Check for too many connected devices – Disconnect devices that are not being used and test the connection again.

11. Update software/drivers – Update the computer's Wi-Fi driver and operating system if only one device is experiencing problems.

12. Test with Ethernet – Connect a computer directly to the router using an Ethernet cable. If Ethernet is fast but Wi-Fi is slow, the problem is likely related to the Wi-Fi connection rather than the ISP.

Q 4-You receive a complaint that a user cannot access Instagram and YouTube on their laptop, but WhatsApp Web works fine. Write down the first 5 troubleshooting steps you would perform to identify if the issue is with hardware, software, or network configuration.

Troubleshooting Steps

If WhatsApp Web works but Instagram and YouTube don't, I would perform these first 5 checks:

1. Check Internet connectivity

- Open Command Prompt and run:

ping 8.8.8.8

- If it works, the laptop has basic Internet connectivity.

2. Test the websites in another browser

- Try Instagram and YouTube in Chrome, Edge, or Firefox.
- If they work in another browser, the problem is likely **browser/software-related**.

3. Check DNS resolution

- Run:

nslookup youtube.com

nslookup instagram.com

- If DNS fails, investigate **DNS/network configuration**.

4. Check firewall, proxy, or security software

- Check whether a Windows firewall rule, antivirus, VPN, or proxy is blocking Instagram/YouTube.
- In Windows, check **Settings → Network & Internet → Proxy**.

5. Test another device on the same Wi-Fi

- Try Instagram and YouTube from a phone connected to the same Wi-Fi.
- **Works on phone but not laptop:** likely laptop/software/configuration issue.
- **Fails on all devices:** likely router, DNS, ISP, or network configuration issue.